TEXORA · LECTURE SHEET

Oxidation, Degradation, Corrosion and Composites

How materials die, and how engineers answer: the electrochemistry of rusting, the corrosion family from galvanic to stress-corrosion cracking, protection by coatings, cathodic methods and alloy design — then the composite idea and the rule of mixtures.

ME-1207 · ENGINEERING MATERIALS · TOPIC 4 · LEVEL 1 TERM 2

1. Corrosion is electrochemistry

Wet corrosion is a short-circuited battery. At **anodic** sites metal dissolves, $Fe \rightarrow Fe^{2+} + 2e^-$; at **cathodic** sites the electrons are consumed, in aerated water by $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$. Iron needs *both* water and oxygen to rust: Fe^{2+} meets OH^- to give $Fe(OH)_2$, oxidised onward to hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$ — **rust**. Rust is porous and non-adherent, so it never seals the surface; attack continues to perforation. Aluminium shows the opposite trick: its oxide Al_2O_3 is dense, adherent and self-healing (Pilling–Bedworth ratio ≈ 1.28 — the oxide occupies slightly more volume than the metal it replaces, so it covers without cracking), which is why aluminium “oxidises” instantly yet survives for decades. An oxide that protects is called **passive**; chromium’s does the same service inside stainless steel (Topic 3).

2. The corrosion family

FORM	PATTERN / TRIGGER
Uniform	Even thinning — predictable, designable (corrosion allowance)
Galvanic	Two dissimilar metals in contact in an electrolyte: the more anodic one corrodes faster — small anode + large cathode is the dangerous geometry
Pitting	Tiny deep holes where chlorides rupture passive films (stainless in seawater) — little weight loss, sudden leaks
Crevice	Stagnant electrolyte under gaskets, washers, lap joints starves oxygen and acidifies — attacks hidden faces
Intergranular	Grain boundaries depleted (sensitised stainless: Cr carbides at boundaries) corrode preferentially — the part can crumble like sugar

Stress-corrosion cracking	Tensile stress + specific corrosive medium (brass in ammonia, steel in caustic) grows cracks at stresses far below yield

Degradation is the wider word: polymers age too — ultraviolet light snaps chains (nylon and polypropylene lose strength outdoors), oxygen attacks at temperature (oxidative scission), solvents swell and plasticise. A fibre, a rope and a pressure pipe all keep calendars, not just load limits.

3. Fighting back

- **Barrier coatings:** paint, enamel, plastic — and **galvanising**, zinc on steel, which is barrier *and* back-up: scratched bare, the more anodic zinc becomes a sacrificial anode and corrodes in the steel's place.
- **Cathodic protection:** force the structure to be the cathode — bolt on sacrificial zinc/magnesium blocks (ship hulls, jetties, buried pipelines) or drive current from a power supply (impressed-current systems).
- **Alloying:** stainless steels, weathering steels (a tight rust that protects), nickel alloys — pay in the metal, save in maintenance.
- **Design and chemistry:** avoid crevices and water traps, insulate dissimilar-metal joints, match the galvanic series, dose inhibitors into closed water systems, and dry or dehumidify stores. In a dyehouse, add: respect chloride limits on stainless vessels and rinse what the process salts.

4. Composites — designed materials

A **composite** combines a **reinforcement** (strong, stiff fibres or particles) with a **matrix** (the glue that holds shape, transfers load and shields the fibres): GFRP and carbon-epoxy (polymer matrix), metal-matrix and ceramic-matrix families, and nature's own — wood, bone, and every woven fabric in resin that a textile engineer will meet in technical textiles. For continuous aligned fibres loaded along their length, the **rule of mixtures** (equal strain) gives

$$E_c = E_f V_f + E_m V_m$$

Glass fibre ($E_f = 72$ GPa) at $V_f = 0.5$ in epoxy ($E_m = 3$ GPa): $E_c = 72 \times 0.5 + 3 \times 0.5 =$ **37.5 GPa**. Across the fibres (equal stress) the soft matrix dominates: $1/E_c = V_f/E_f + V_m/E_m$ gives only **5.8 GPa** — the same material, six times weaker across the grain. That anisotropy is not a flaw but a design variable: orient plies where the loads run, exactly as a weaver chooses warp and weft. On a weight basis the composite embarrasses metals: E/ρ for the GFRP $\approx 37.5/1.9 \approx 20$, for steel $200/7.85 \approx 25$ — comparable stiffness per kilo, with far better corrosion behaviour; carbon fibre ($E = 230$ GPa, $\rho_{composite} \approx 1.6$) reaches $E/\rho \approx 87$ along the fibres, which is why aircraft and racing machines are black.

5. Key points

- Rusting needs water + oxygen; anode dissolves metal, cathode consumes electrons; rust does not protect, alumina and chromia do (P–B ≈ 1.28 for Al_2O_3).
- Galvanic, pitting, crevice, intergranular and stress-corrosion attack are local and treacherous; uniform corrosion is the honest one.
- Defences: barriers, galvanising (barrier + sacrificial), cathodic protection, corrosion-resistant alloys, trap-free design.
- Composites: E_c = E_fV_f + E_mV_m along the fibres; transverse stiffness collapses toward the matrix value — design the orientation.

6. Practice

Answer in your own words; brief model answers follow.

1. A unidirectional carbon-fibre/epoxy laminate has V_f = 0.6 (E_f = 230 GPa, E_m = 3 GPa). Find the longitudinal and transverse Young's moduli.
2. A galvanised sheet is scratched through to the steel. Explain why the exposed steel still resists rusting, naming the two protection mechanisms of the zinc coat.
3. Steel screws fasten a copper plate on a boat hull below the waterline. Predict the corrosion outcome and the design error, referencing anode/cathode areas.
4. Iron left outdoors disintegrates over years while an aluminium sheet nearby merely dulls. Explain both behaviours through their oxides.
5. Why does a woven (0°/90°) fabric laminate show more balanced properties than a unidirectional one, and what does it sacrifice?
6. A stainless dyeing vessel develops cracks along welds after a year of chloride-bearing baths. Name the likely corrosion forms involved and two remedies.

Answer key

1. Longitudinal: E_c = 230 × 0.6 + 3 × 0.4 = 138 + 1.2 = 139 GPa. Transverse: 1/E_c = 0.6/230 + 0.4/3 = 0.0026 + 0.1333 = 0.1359 → E_c = 7.4 GPa. Nearly twentyfold anisotropy — lay-up design is not optional.
2. Intact, the zinc is a barrier coat. At the scratch, zinc and steel form a galvanic couple in the moisture film; zinc is the more anodic metal, so it corrodes preferentially as a sacrificial anode while the steel becomes the protected cathode. The zinc is consumed slowly around the scratch, but the steel waits — protection with a bodyguard.

3. The steel screws corrode rapidly. Steel is anodic to copper in seawater, and the geometry is the worst case: tiny anodes (screws) feeding a huge cathode (plate) concentrate all the galvanic current into the small parts, which eat through quickly. Correct design matches metals, insulates the joint, or makes the fastener the nobler (cathodic) material.
4. Aluminium’s oxide is dense, adherent, self-repairing and slightly voluminous (P–B ratio ≈ 1.28), so it seals the metal after a first invisible tarnish. Iron’s rust is porous, cracked and loosely attached: water and oxygen pass freely, flakes spall off, fresh metal is exposed — the “oxide” accelerates nothing but offers no defence, so section loss proceeds until perforation.
5. A 0°/90° weave splits the fibres between the two main directions, so stiffness and strength along warp and weft are comparable rather than one superb and one poor direction. The sacrifice: peak longitudinal performance halves roughly (only half the fibres serve each direction) and crimped, wavy woven fibres are less efficient than straight ones.
6. Likely a partnership: sensitisation from welding precipitated chromium carbides, enabling intergranular attack, aggravated by chloride pitting/crevice corrosion in the bath deposits (and possibly stress-corrosion cracking under residual weld stress). Remedies: use low-carbon (304L/316L) or stabilised grades and solution-anneal after welding; upgrade to a molybdenum-bearing grade (316) for chloride duty, control bath chlorides and rinse/passivate the vessel on schedule.

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